

Encryption, Decryption, Hashing

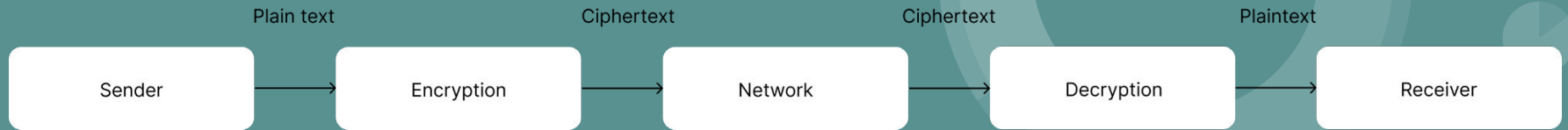
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Why need encryption?



Network security



Encryption

Symmetric key encryption

Asymmetric key encryption

Encryption

(used to protect sensitive information)



Plain text



Encryption



Encrypted text



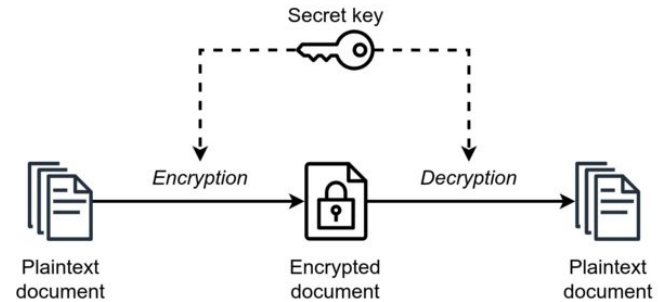
Decryption



Plain text

Symmetric key encryption

Symmetric encryption uses one key to encrypt and decrypt. If you encrypt a zip file, and then decrypt with the same key, you are using symmetric encryption. Symmetric encryption is also called “secret key” encryption, as the key must be kept secret from third parties.



Symmetric key encryption



Symmetric key encryption



128 and 256 bit

Block Cipher



64 and 128 bit

- DES
- AES
- Triple DES
- Blowfish
- O'z DSt 1105:2009

Stream Cipher



8 bit

- RC4
- Salsa 20
- Panama

Diffie Hellman key exchange

AES

		Y															
		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	63	7C	77	7B	F2	68	6F	C5	30	01	67	2B	FE	D7	AB	76	
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	CO	
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15	
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75	
4	09	83	2C	1A	1B	6E	5A	A0	52	38	D6	B3	29	E3	2F	84	
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF	
6	DO	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8	
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2	
8	CD	OC	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73	
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	OB	DB	
a	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79	
b	E7	C8	37	6D	8D	05	4E	A9	6C	56	F4	EA	65	7A	AE	08	
c	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	48	BD	8B	8A	
d	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E	
e	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF	
f	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16	

Figure 3. AES S-box

The inverse operation, called InvSubBytes, consists of the use of the inverse S-Box of fig. 4 for each byte of the state.

		Y															
		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	52	09	6A	D5	30	36	A5	38	BF	40	A3	9E	81	F3	D7	FB	
1	7C	E3	39	82	9B	2F	FF	87	34	83	43	44	C4	DE	E9	CB	
2	54	7B	94	32	A6	C2	23	3D	EE	4C	95	08	42	FA	C3	4E	
3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1	25	
4	72	F8	F6	64	86	68	98	16	D4	A4	5C	CC	5D	65	B6	92	
5	6C	70	48	50	FD	ED	B9	DA	5E	15	46	57	A7	8D	9D	84	
6	90	D8	AB	00	8C	BC	D3	0A	F7	E4	58	05	B8	B3	45	06	
7	D0	2C	1E	8F	CA	3F	0F	02	C1	AF	BD	03	01	13	8A	68	
8	3A	91	11	41	4F	67	DC	EA	97	F2	CF	CE	F0	B4	E6	73	
9	96	AC	74	22	E7	AD	35	85	E2	F9	37	EB	1C	75	DF	6E	
a	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE	1B	
b	FC	56	3E	4B	C6	D2	79	20	9A	DB	CO	FE	78	CD	5A	F4	
c	1F	DD	A8	33	88	07	C7	31	B1	12	10	59	27	80	EC	5F	
d	60	51	7F	A9	19	B5	4A	0D	2B	E5	7A	9F	93	C9	9C	EF	
e	A0	E0	3B	4D	AE	2A	F5	B0	CB	EB	BB	3C	83	53	99	61	
f	17	2B	04	7E	BA	77	D6	26	E1	69	14	63	55	21	OC	7D	

Figure 4. AES Inverse S-box

ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]	48	30	110000	60	0	96	60	1100000	140	.
1	1	1	1	[START OF HEADING]	49	31	110001	61	1	97	61	1100001	141	a
2	2	10	2	[START OF TEXT]	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	[END OF TEXT]	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	[END OF TRANSMISSION]	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	[ENQUIRY]	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	[ACKNOWLEDGE]	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	[BELL]	55	37	110111	67	7	103	67	1100111	147	g
8	8	1000	10	[BACKSPACE]	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	[HORIZONTAL TAB]	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	[LINE FEED]	58	3A	111010	72	:	106	6A	1101010	152	j
11	B	1011	13	[VERTICAL TAB]	59	3B	111011	73	;	107	6B	1101011	153	k
12	C	1100	14	[FORM FEED]	60	3C	111100	74	<	108	6C	1101100	154	l
13	D	1101	15	[CARRIAGE RETURN]	61	3D	111101	75	=	109	6D	1101101	155	m
14	E	1110	16	[SHIFT OUT]	62	3E	111110	76	>	110	6E	1101110	156	n
15	F	1111	17	[SHIFT IN]	63	3F	111111	77	?	111	6F	1101111	157	o
16	10	10000	20	[DATA LINK ESCAPE]	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	[DEVICE CONTROL 1]	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	[DEVICE CONTROL 2]	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	[DEVICE CONTROL 3]	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	[DEVICE CONTROL 4]	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	[SYNCHRONOUS IDLE]	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	[END OF TRANS. BLOCK]	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	[CANCEL]	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	[END OF MEDIUM]	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	[SUBSTITUTE]	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	[ESCAPE]	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	[FILE SEPARATOR]	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	[GROUP SEPARATOR]	77	4D	1001101	115	M	125	7D	1111101	175	}
30	1E	11110	36	[RECORD SEPARATOR]	78	4E	1001110	116	N	126	7E	1111110	176	~
31	1F	11111	37	[UNIT SEPARATOR]	79	4F	1001111	117	O	127	7F	1111111	177	[DEL]
32	20	1000000	40	[SPACE]	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	{	88	58	1011000	130	X					
41	29	101001	51	}	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

AES

Textni 16 baytli yani 128 bitli bo'laklarga bo'linadi

Key tanlanadi 128 192 256 bitli

Round key qo'shiladi

S Box ustinigga baytlar almashtiriladi

Qatorni siljitish $n-1$ ga chapga

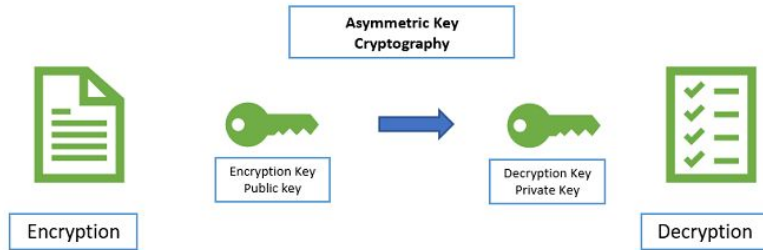
Oldindan belgilangan const
matrixga kopaytiramiz

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

Constant Matrix

Asymmetric key encryption

- Public key
- Private key



Asymmetric key encryption

- RSA
- Elgamal
- DSA
- Ed25519



Symmetric key encryption

It only requires a single key for both encryption and decryption.

The size of cipher text is the same or smaller than the original plain text.

The encryption process is very fast.

The length of key used is 128 or 256 bits

Asymmetric key encryption

It requires two keys, a public key and a private key, one to encrypt and the other one to decrypt.

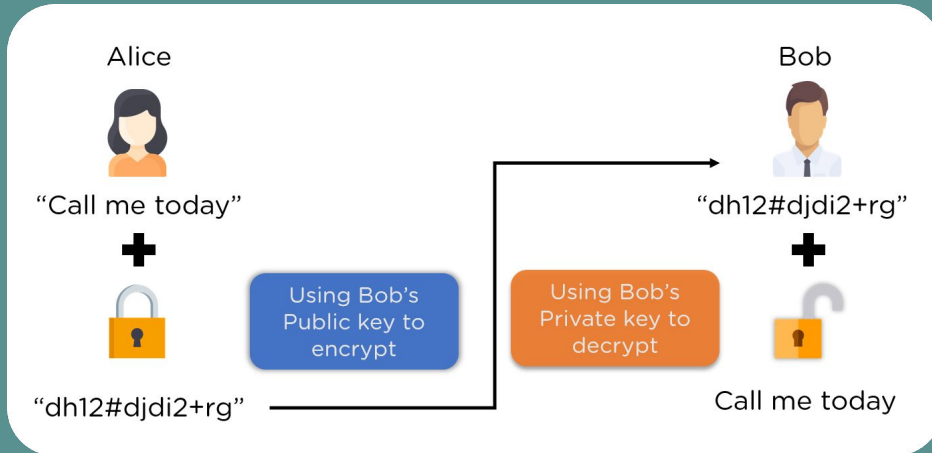
The size of cipher text is the same or larger than the original plain text.

The encryption process is slow.

The length of key used is 2048 or higher

RSA(Rivest-Shamir-Adelman)

- Public key encryption
- Private key decryption



RSA(Rivest-Shamir-Adelman)

2 ta katta tup sonlar tanlab olinadi P va Q

n ni topamiz $= P * Q$

Encryption key tanlanadi(Public key)

$1 < e < F(n)$ va $GCD(e, F(n)) = 1$

F(n) topiladi $F(n) = F(P) * F(Q)$

Tub sonlar uchun $F(P) = P-1$

Decryption key topiladi(Private key):

$$d = e^{-1} \bmod F(n)$$



$$d * e = 1 \bmod F(n)$$



$$d * e * \bmod F(n) = 1$$

d lar soni cheksiz

Encryption

$$c = m^e \bmod n$$

c = ciphertext

m = plaintext

$m < n$

Decryption

$$m = c^d \bmod n$$

c = ciphertext

m = plaintext

$m < n$

Diffie Hellman key exchange

q tub soni tanlanadi

a soni tanlanadi

$a^1 a^2 \dots a^{q-1} \bmod q \in \{1, 2, 3, \dots, q-1\}$



Private keyni tanlaymiz $X(a) < q$

Public keyni topamiz

$Y(a) = a^{X(a)} \bmod q$

Huddi shunday usul bilan $X(b)$ Private keyni tanlaymiz $X(b) < q$

Public keyni topamiz

$Y(b) = a^{X(b)} \bmod q$



$k1$ ni topamiz $k1 = Y(b)^{X(a)} \bmod q$

$k2$ ni topamiz $k2 = Y(a)^{X(b)} \bmod q$

Taqqoslaymiz $k1=k2$

Elgamal

P bitta katta tub son tanlanadi

Private key tanlanadi $\rightarrow d$

Birinchi encryption key tanlanadi $\rightarrow e(1)$

Ikkinchi encryption key topiladi $\rightarrow e(2)$

$$e(2) = e(1)^d \mod P$$

Bitta int son tanlanadi $\rightarrow R$

$$c(1) = e(1)^R \mod P \rightarrow \text{CipherText}$$

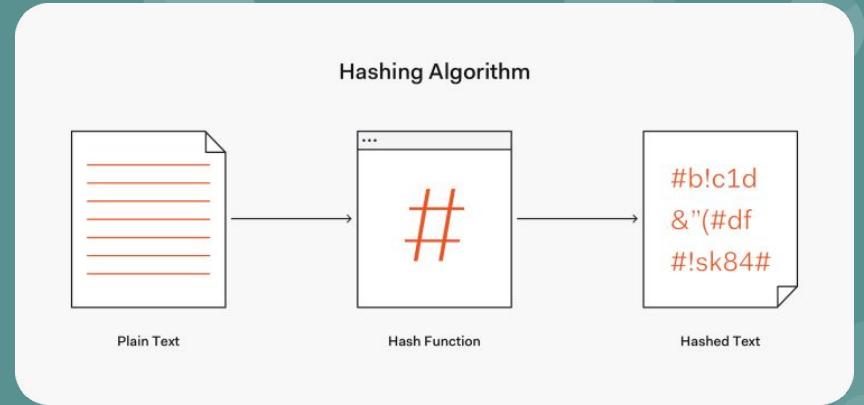
$$c(2) = P(t)^R e(2) \mod P \rightarrow \text{CipherText}$$

$P(t)$ plaintext

$$P(t) = c(2) * (c(1))^{-d} \mod P$$

Hashing

Hashing is the practice of transforming a given key or string of characters into another value for the purpose of security. Unlike standard encryption, hashing is always used for one-way encryption, and hashed values are very difficult to decode.



Hashing

- MD5
- SHA(1,2,128,256)



MD5



128 bit

- Padding bits
- Appending
- Padding length
- Initialize MD Buffer
- Process each block



Hash collision

MD5: The fastest and shortest generated hash (16 bytes). The probability of just two hashes accidentally colliding is approximately: $1.47 \cdot 10^{-29}$.

SHA1: Is generally 20% slower than md5, the generated hash is a bit longer than MD5 (20 bytes). The probability of just two hashes accidentally colliding is approximately: $1 \cdot 10^{-45}$

SHA256: The slowest, usually 60% slower than md5, and the longest generated hash (32 bytes). The probability of just two hashes accidentally colliding is approximately: $4.3 \cdot 10^{-60}$.